Roll a die 6 times.
What is the probability that all the 6 outcomes appear?

$$p = \frac{6!}{6^6}$$

The same question but with 7 rolls.

$$q = \frac{6 \cdot \frac{7!}{2}}{6^7}$$

1 1 2 3 4 5 6
1 2 2 3 4 5 6
⋮

The same question with 100 rolls.

≈ 1.

$$p = \frac{k}{n}$$

$$p = \frac{3}{6}$$

$$\Omega = \{1,2,3,4,5,6\}$$

$\underbrace{A\ \bar{A}\ \cdots\ A}_{n \text{ trials}}$

k_A times happened A

k_A : frequency of A

$\frac{k_A}{n}$: relative frequency of A

$$\frac{k_A}{n} \longrightarrow \mathbb{P}(A)$$

i) $\frac{k_A}{n} \geq 0$ ii) $\frac{k_\Omega}{n} = 1$

iii) If $A_1, A_2, \dots$ are mutually exclusive events then

$$\frac{k_{A_1 \cup A_2 \cup \dots}}{n} = \frac{k_{A_1}}{n} + \frac{k_{A_2}}{n} + \dots$$

i)* $\mathbb{P}(A) \geq 0$ ii)* $\mathbb{P}(\Omega) = 1$

iii)* If $A_1, A_2, \dots$ are mutually exclusive events then

$$\mathbb{P}(A_1 \cup A_2 \cup \dots) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \dots$$

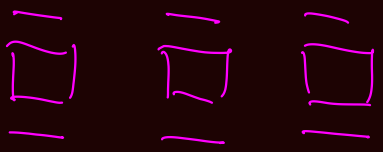
Nikolai Kolmogorov (1936)

$(\Omega, \mathbb{P})$ $\mathbb{P}(A) \in [0,1]$

A is an event if $A \subset \Omega$.

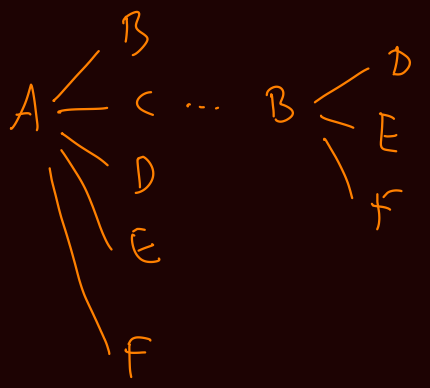
Ex. 6 people are playing chess.

A, B, C, D, E, F.



What is the probability that A plays with B
C " " D
E " " F

$$\frac{3! \cdot 2^3}{6!} = \frac{2^3}{5!} = \frac{2 \cdot 2 \cdot 2}{2 \cdot 3 \cdot 4 \cdot 5} = \frac{1}{3 \cdot 5}$$



5·3 possibilities

$$P = \frac{1}{3 \cdot 5}$$

$$\mathbb{P}(\emptyset) = 0$$

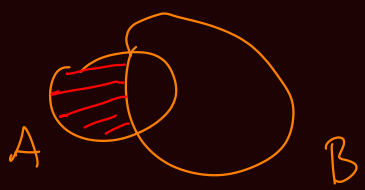
proof $\mathbb{P}(\underbrace{\emptyset \cup \emptyset \cup \cdots}_{\emptyset}) = \mathbb{P}(\emptyset) + \mathbb{P}(\emptyset) + \cdots$

$$\mathbb{P}(\emptyset) = \mathbb{P}(\emptyset) + \mathbb{P}(\emptyset) + \cdots$$

$$\mathbb{P}(\emptyset) = 0$$

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

$$\mathbb{P}(A - B) = \mathbb{P}(A) - \mathbb{P}(B) \quad \text{if} \quad A \supset B$$



$A = \Omega$

$$\mathbb{P}(\bar{B}) = 1 - \mathbb{P}(B).$$

$$\mathbb{P}(A_1 \cup A \cup \cdots) \leq \mathbb{P}(A_1) + \mathbb{P}(A_2) + \cdots$$

Ex. We roll a die 10 times

A: the maximum is 4.

$\mathbb{P}(A) = ?$

Sol 1. $\frac{2}{6^{10}}$

Sol 2 B: the maximum is at most 4

C: " " " " " 3

A: the max $= 4$

$A = B - C$ and $B \supset C$

$\mathbb{P}(A) = \mathbb{P}(B) - \mathbb{P}(C) = \frac{4^{10}}{6^{10}} - \frac{3^{10}}{6^{10}}$.

Conditional probability

Suppose the B has happened.

$\mathbb{P}(A|B)$: the prob. of A conditional on B.

A: the die shows 5

B: the die shows an even number

$\mathbb{P}(A|B) = \frac{1}{3}$

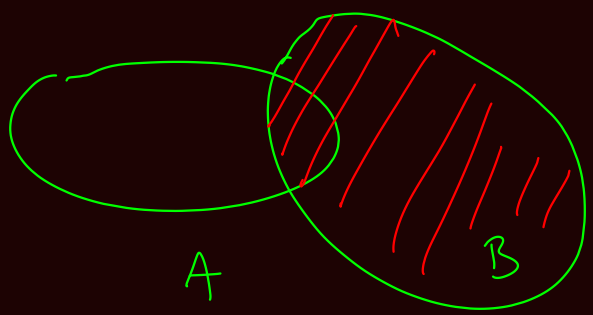
$\underbrace{A\ \bar{A}\ \cdots\ A}_{n \text{ times}}$

$\frac{k_A}{n} \approx \mathbb{P}(A)$ if n is "huge".

Suppose that B has happened

$$\frac{k_{A\cap B}}{k_B} \approx \mathbb{P}(A|B)$$

$$\frac{k_{A\cap B}/n}{k_B/n} \approx \frac{\mathbb{P}(A\cap B)}{\mathbb{P}(B)} \rightsquigarrow \mathbb{P}(A|B) = \frac{\mathbb{P}(A\cap B)}{\mathbb{P}(B)}, \text{ if } \mathbb{P}(B)>0.$$



$\frac{\mathbb{P}(A\cap B)}{\mathbb{P}(B)}$

A: the roll is 6

B: the roll is even

$$\mathbb{P}(A|B) = \frac{1/6}{3/6} = \frac{1}{3}.$$